ONUMA x TA-14 / LT-2 EDUCATIONAL TECHNICAL EXAMINATION

The Building Knows. Who Authorizes Reality?

Semantic continuity, admissible computation, and the physical consequence boundary

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The question is not whether a digital building can identify an asset or propose an action. It is whether the evidence supporting a proposed physical consequence is current, attributable, sufficiently complete, and within applicable authority.

Controlling distinctions

Representation is not reality. A model describes an asset relationship; it does not establish current physical condition. Prediction is not permission. A computed result may guide investigation without independently authorizing intervention. Assignment is not authorization. Being named in a work order does not establish permission for every physical act.

Scope and attribution

This is a TA-14-authored educational examination referring to an externally generated ONUMA / BIMgenie operational case. It is not a jointly approved standard, proof of deployed software integration, certification, or field validation. Four illustrations are conceptual. The HVAC illustration is separate from the LT-2 electrical case.

Reference: BIMgenie Work Order 3593-22790; RE-7, Living Room Small; reported non-illuminating light and breaker appearing ON. The existing public LT-2 case records supported represented topology, bounded investigation allowed, and physical electrical intervention held.

01 / Executive examination

The proposition

A persistent semantic representation can make building knowledge more usable across design, construction, operation, modification, and transfer. But semantic continuity does not itself verify present physical state or confer authority to act.

What the referenced LT-2 case supports

The recorded ACA examination supports a bounded represented electrical relationship associated with LT-2. The AEA result allows identification, accessible observation, comparison of visible labels, discrepancy preservation, and return of new evidence. It does not authorize electrical operation, access, measurement, switching, repair, or closure.

What remains unestablished

The physical affected light has not been independently verified as LT-2; the panelboard identifier relationship has not been physically confirmed; the reported breaker position is not a verified energized state; and the cause of the non-illuminating light is unknown.

Decision rule

For each proposed consequence, distinguish (1) what the record represents, (2) what current field evidence establishes, (3) who has authority and standing, and (4) what bounded action follows. An incomplete chain yields a HOLD or another explicitly justified determination, not invented certainty.

Independent review standard

An examiner should be able to identify the source object, reconstruct the claimed relationship from available evidence, point to every unresolved physical-state claim, and explain why a permitted investigation does not imply permission to intervene.

02 / Architecture: distinct responsibilities

Layer	Primary question	Boundary
Semantic Bridge / information continuity	Which assets, spaces, documents, people, and operational records refer to the same building reality?	Preserves meaning and relationships; does not verify physical truth by itself.
ACA / admissible computation	What proposition is supported by the supplied computation and evidence, within its stated scope?	A supported model relationship is not a field measurement.
AEA / admissible execution	Does this specific proposed consequence have Admissible Evidence, Applicable Authority, and Established Standing now?	May ALLOW, HOLD, DENY, or ESCALATE; authorization is bounded.
Post-action verification	What changed, and is the outcome established by attributable observations?	Completion status alone is not outcome evidence.

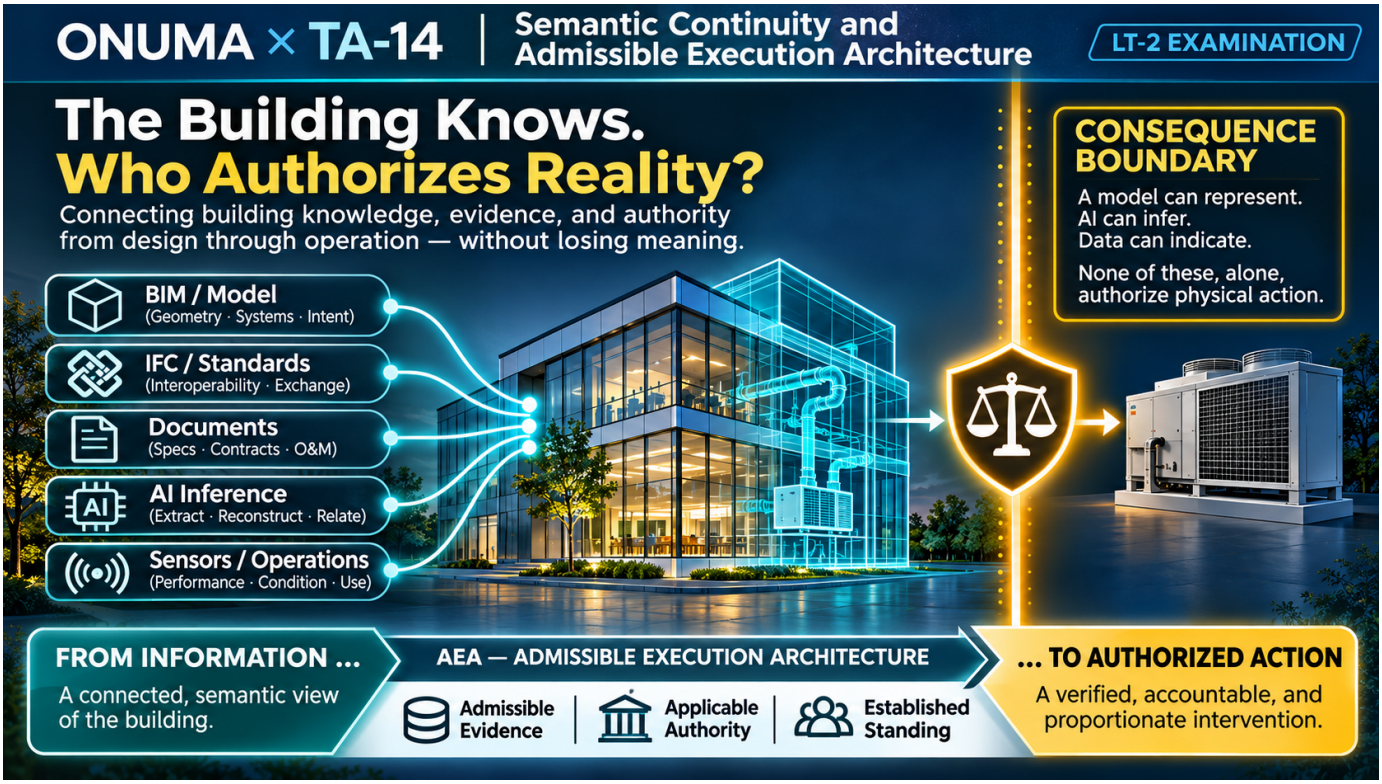
No implied technical integration

These layers are presented as complementary conceptual responsibilities. This document does not claim that ONUMA, BIMgenie, a Semantic Bridge, ACA, and AEA are deployed as one interoperable production stack.

A proposed consequence is an explicit object

A responsible decision identifies the target physical asset, requested operation, purpose, expected effect, actor, authority basis, relevant present conditions, limits, and evidence expiry or revalidation triggers. If those fields are missing, a model-based recommendation must not silently become a command.

03 / Illustration: building knowledge to authorized action



Reading the diagram

BIM, IFC, documents, AI inference, and operational signals converge into a contextual view. The gold boundary represents a separate decision about the admissibility of physical action, not a claim that the upstream systems themselves grant permission.

Examiner prompt: Which source established the exact asset identity, which observation is current, and which party is authorized to execute the proposed operation? If any answer depends solely on inference, document the gap.

04 / Illustration: the information continuity chain

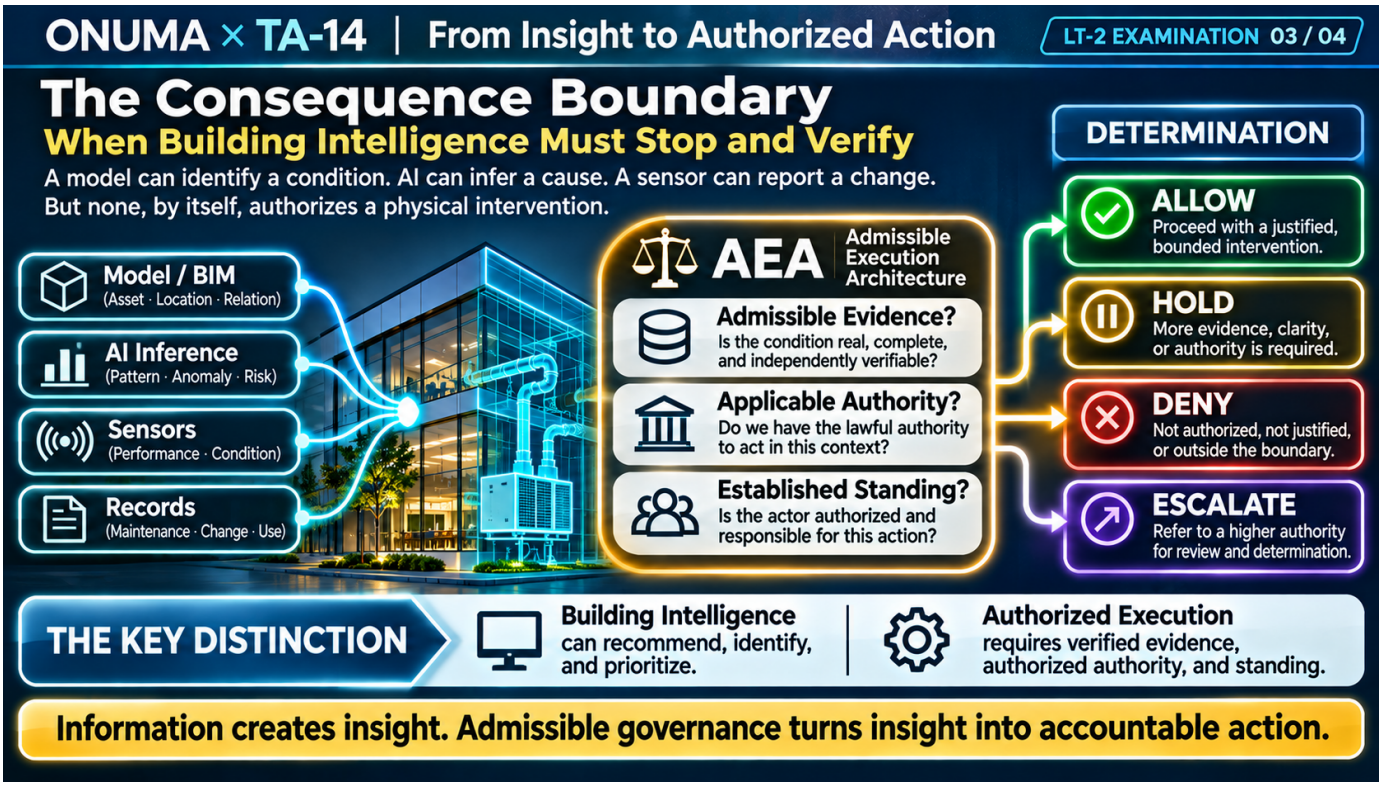


What continuity should preserve

Design intent; as-built deviations; commissioning evidence; maintenance and operating observations; later modifications; asset identifiers; the provenance of inferred relationships; and unresolved discrepancies at handoff.

Failure mode: A perfectly connected but outdated model can be more persuasive than a disconnected record while remaining wrong. Traceable change and current-state verification are therefore separate requirements.

05 / Illustration: the consequence boundary

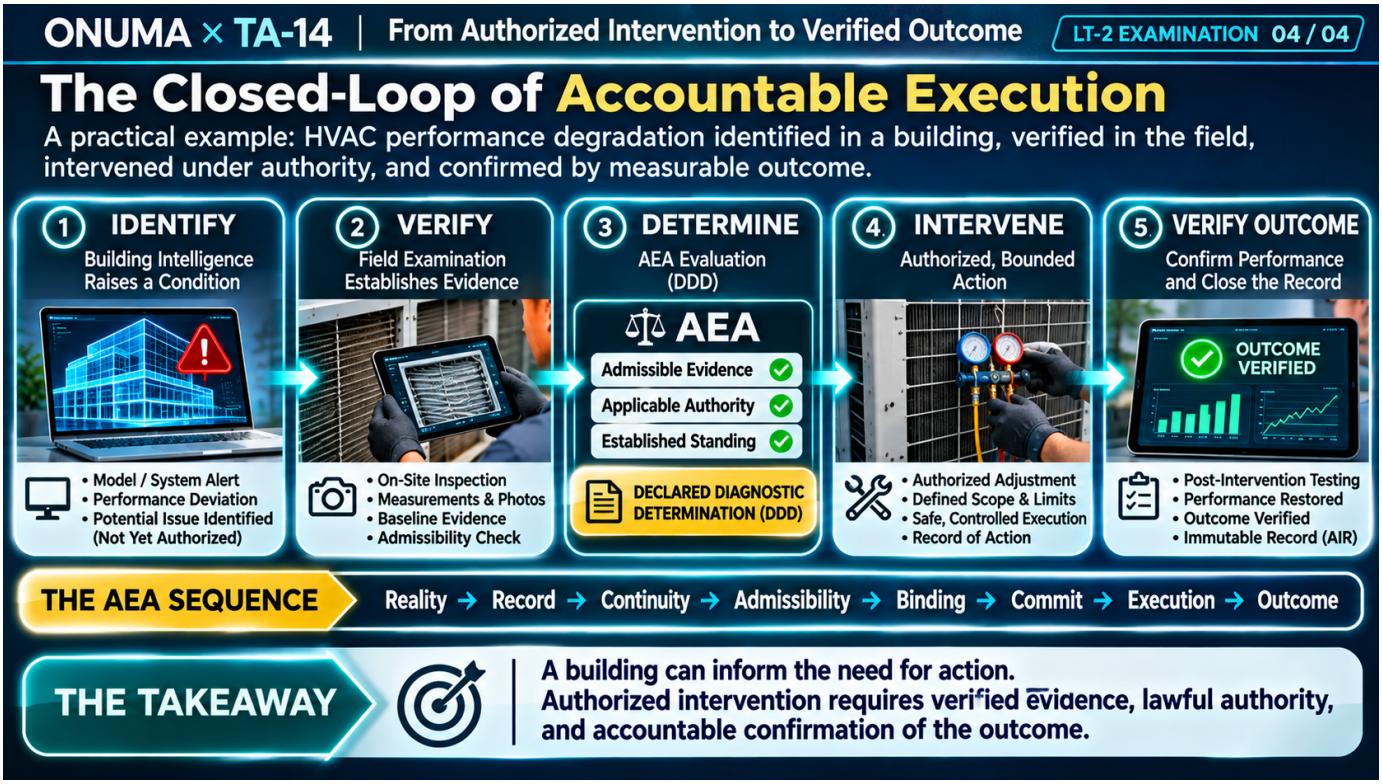


Four bounded determinations

ALLOW: Proceed only within evidenced scope and authority. HOLD: Preserve the boundary pending missing evidence, context, or authorization. DENY: Reject a proposed act outside the admissible boundary. ESCALATE: Route a decision to an appropriate responsible authority.

The illustration is a conceptual explanation of AEA, not a result from a live command-control system. For LT-2, the controlling distinction remains ALLOW bounded non-intervention investigation and HOLD physical electrical intervention.

06 / Illustration: accountable execution and outcome



Separate illustrative HVAC workflow

An alert can initiate guided field examination. A technician gathers authentic physical observations and completes the required baseline sequence. A system-issued Declared Diagnostic Determination (DDD) follows the completed, frozen baseline; it is not authored or edited by the technician.

A DDD establishes the diagnostic determination and justified corrections, including no intervention when warranted. AEA separately evaluates whether a proposed intervention is authorized. Where intervention occurs, post-intervention evidence is collected and the outcome is assessed. This HVAC example is not evidence of any action in the LT-2 electrical case.

07 / LT-2: preserved evidence and missing reality

Reference object: BIMgenie Work Order 3593-22790. Location: RE-7, Living Room Small. Trade: Electrical. Operational state: Work In Progress. The occupant report describes a light that is not illuminating and a local Circuit 2 breaker that appears ON.

Proposition	Status in public case	Permitted inference
Reported lighting condition	Reported, not independently field-verified	Justifies bounded investigation.
LT-2 represented topology	Supported by bounded ACA examination	Supports represented relationship only.
Physical affected light = LT-2	Not established	Must identify in the field.
Panelboard V03 = physical DB-2	Not independently established	Compare accessible identifiers; preserve mismatch.
Breaker actually energized	Not established; “appears ON” is a report	No switching or electrical measurement authorized.
Cause and correction	Not established	No repair or replacement authorized.

Interpretation

The central result is not a failed repair. It is the preservation of the boundary between a supported digital relationship and an unverified physical state. A valid examiner may agree with HOLD while still questioning the quality or provenance of the represented topology evidence.

08 / Evidence-to-consequence matrix

Proposed action	Current status	Evidence or authority still needed
Locate assigned room; observe accessible conditions	ALLOW: bounded investigation	Remain non-invasive; record time and observed object.
Compare visible labels to model	ALLOW: bounded investigation	Preserve disagreements without forcing equivalence.
Operate breaker or access panel	HOLD	Physical identification, current-state evidence, and applicable authorization.
Perform electrical measurement	HOLD	Qualified actor, authorized scope, safe procedure, and evidence.
Repair, replace, or rewire	HOLD	Verified condition, justified correction, authorization, and standing.
Close work order as resolved	HOLD	Appropriate evidence of resolution and closure authority.

Why this is not a universal permission table

These statuses summarize the particular public LT-2 examination. They are not general electrical safety rules, licenses, or a substitute for site-specific procedures. New evidence or changed context requires a new determination, not an automatic transition from HOLD to ALLOW.

09 / Falsification, revalidation, and review

Conditions that could undermine a represented finding

Conflicting asset identifiers, revised topology, an undocumented equipment replacement, a mismatch between room and physical device, or an untraceable source transformation would require the represented claim to be reconsidered.

Evidence needed before considering physical intervention

A clearly identified target; current, attributable field observations; any necessary safe and lawful inspection or testing performed under proper authority; a defensible condition and correction; the responsible actor and their standing; and a fresh AEA evaluation of the proposed act. The public record does not show these requirements completed.

Independent examination challenge

1. Reproduce or independently scrutinize the bounded topology proposition using the source evidence if made available. 2. Trace the provenance of every material identifier. 3. Enumerate unresolved physical claims. 4. Explain why non-invasive investigation is permitted but intervention is held. 5. State what new observations would alter the determination. 6. Record disagreements and limits without converting uncertainty into a PASS.

Review limitation

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10 / Conclusions, controls, and references

Conclusions

A building information system may preserve meaning across applications and decades. That is valuable but insufficient for physical authorization. AEA addresses the separate question of whether a specific proposed consequence is justified under current evidence, applicable authority, and established standing. The LT-2 case illustrates why a defensible HOLD is a meaningful result.

Publication controls

Publisher: TA-14 / independent educational examination. Version: 2.0. Date: 09 November 2026. Scope: conceptual examination and commentary on the existing public LT-2 case. Not established: deployed ONUMA-TA-14 integration, joint endorsement, third-party certification, or successful field intervention.

Public source and further examination

ONUMA / BIMgenie LT-2 public case:
https://www.ta14exchange.com/ai-governance/admissible-architecture/onuma-lt2

TA-14 AEA registry:
https://www.ta14exchange.com/registry/ta-14-admissible-execution-architecture

This document reflects the public case as supplied for the examination. It does not independently authenticate the underlying operational record.